

IGCSE Chemistry CIE

YOUR NOTES



12. Experimental Techniques & Chemical Analysis

CONTENTS

- 12.1 Experimental Techniques
 - 12.1.1 Apparatus for Measurements
 - 12.1.2 Solutions
 - 12.1.3 Acid-Base Titrations
- 12.2 Separation & Purification
 - 12.2.1 Paper Chromatography
 - 12.2.2 Locating Agents & R_f Values
 - 12.2.3 Separation & Purification Techniques
- 12.3 Identification of Ions & Gases
 - 12.3.1 Identification of Anions
 - 12.3.2 Identification of Cations
 - 12.3.3 Identification of Gases

12.1 Experimental Techniques

12.1.1 Apparatus for Measurements

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Time, Temperature, Mass & Volume

Time

- Time can be measured using a stopwatch or stopclock which are usually accurate to one or two decimal places
- The units of time normally used are seconds or minutes although other units may be used for extremely slow reactions (e.g. rusting)
- 1 minute = 60 seconds

Temperature

- Temperature is measured with a thermometer or digital probe
- Laboratory thermometers usually have a precision of a half or one degree
- Digital temperature probes are available which are more precise than traditional thermometers and can often read to 0.1 °C
- Traditional thermometers rely upon the uniform expansion and contraction of a liquid substance with temperature; digital temperature probes can be just as, if not, more accurate than traditional thermometers
- The units of temperature are **degrees Celsius (°C)**

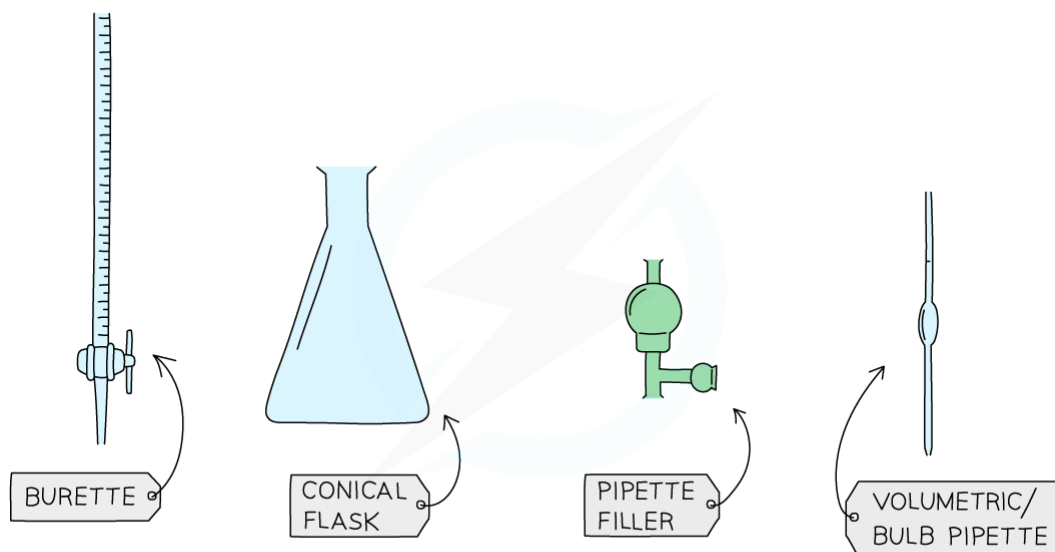
Mass

- Mass is measured using a digital balance which normally gives readings to two decimal places
- Balances must be tared (set to zero) before use
- The standard unit of mass in kilograms (kg) but in chemistry grams (g) are most often used
- 1 kilogram = 1000 grams

Volume-liquids

- The volume of a liquid can be determined using several types of apparatus, depending on the level of accuracy needed
- For approximate volumes where high accuracy is not an important factor, measuring (or graduated) cylinders are used
- These are graduated (have a scale so can be used to measure) and are available typically in a range of sizes from 10 cm³ to 1 litre (1 dm³)
- Volumetric pipettes are the most accurate way of measuring a **fixed** volume of liquid, usually 10 cm³ or 25 cm³
 - They have a scratch mark on the neck which is matched to the bottom of the meniscus to make the measurement
- Burettes are the most accurate way of measuring a **variable** volume of liquid between 0 cm³ and 50 cm³ (e.g. in a titration)
- The tricky thing with burettes is to remember to read the scale from top to bottom as 0.00 cm³ is at the top of the column

- Whichever apparatus you use, you may see markings in ml(millilitre) which is the same as a cm^3



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Diagram of a burette with conical flask and pipette with filler

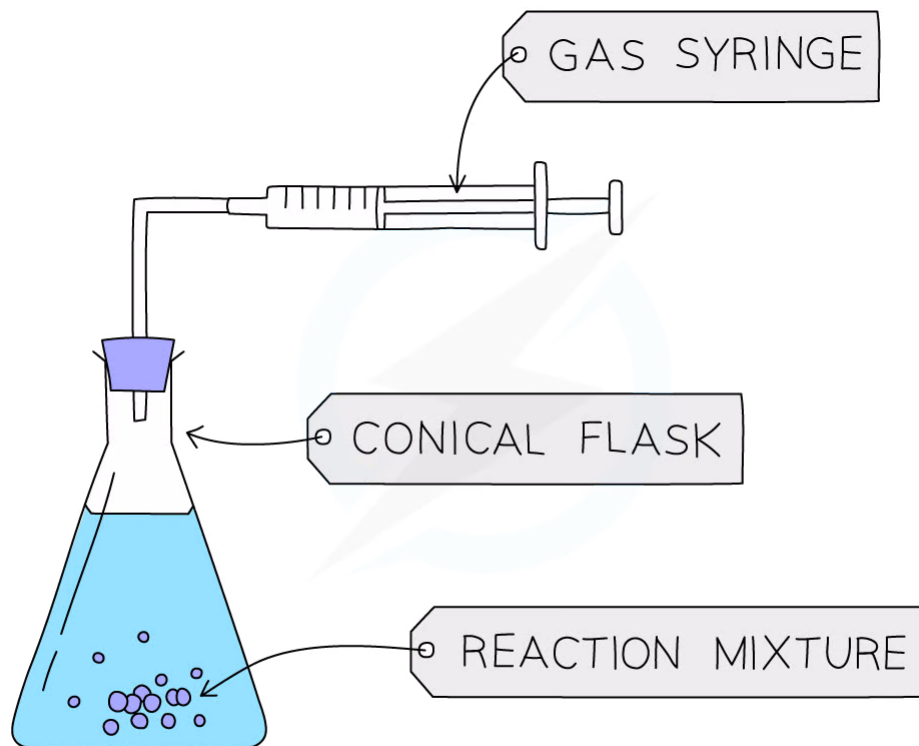
Volume-gases

- The volume of a gas sometimes needs to be measured and is done by collecting it in a graduated measuring apparatus
- A **gas syringe** is usually the apparatus used
- A graduated cylinder inverted in water may also be used, provided the gas isn't water-soluble
- If the gas happens to be heavier than air and is coloured, the cylinder can be used upright

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Diagram of the set-up for an experiment involving gas collection



Exam Tip

Be careful when recording time not to mix up seconds and minutes in the same table. If a table heading shows Time/mins and you record a stop watch display of 1.30, meaning 1 minute and 30 seconds, that is wrong as it should be 1.5 mins. To avoid any confusion, if the time intervals are less than a minute, its best to change the recorded units to seconds. That same stopwatch display would therefore be recorded as 90 seconds.

Advantages & Disadvantages of Methods & Apparatus

- In the lab we often have choices of different apparatus to do the same job
- Evaluating what is the best one to use is part of good experimental planning and design
- This means appreciating some of the advantages and disadvantages of laboratory apparatus

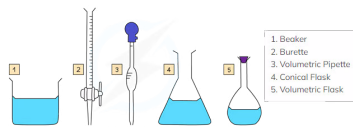
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Table showing examples of Advantages and Disadvantages of Lab Apparatus

Apparatus choice	Advantage	Disadvantage
Temperature probe versus thermometer	◦ More precise readings ◦ Can easily take multiple repeat readings ◦ Can use automatic sampling over longer periods of time	◦ Probe can be corroded by some reagents ◦ Probes more expensive to replace
Volumetric pipette versus a measuring cylinder	◦ Pipette measures very accurately	◦ Harder to use and only measures one fixed volume
Gas syringe versus inverted cylinder for collecting gases	◦ Gas syringes are easy to set up and keep the gas dry	◦ The pistons can stick ◦ Limited volume can be collected ◦ Delicate and expensive
Microscale versus normal scale quantities	◦ Less wasteful of reagents ◦ Saves energy ◦ Safer	◦ Hard to see what's happening ◦ Lose a lot of material in separating and purifying products

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Five pieces of apparatus that can be used to measure the volume of a liquid. They all have their pros and cons

Planning your method

- Good experimental design includes the answers to questions like
 - Have I chosen a suitable apparatus for what I need to measure?
 - Is it going to give me results in an appropriate time frame?
 - Is it going to give me enough results to process, analyse and make conclusions?
 - Does it allow for repetitions to check how **reliable** my results are?
 - Does my plan give a suitable **range** of results?
 - How can I be sure my results are **accurate**?
 - Have I chosen an appropriate scale of quantities without being wasteful or unsafe?
- You may be asked about experimental methods in exam questions and your experience and knowledge of practical techniques in chemistry should help you to spot mistakes and suggest improvements



Exam Tip

Make sure you know the names of common laboratory apparatus

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12.1.2 Solutions

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Solutions

- You need to know all the following terms used when describing solutions:

Terminology About Solutions Table

Term	Meaning	Example
Solvent	The liquid in which a solute dissolves	The water in seawater
Solute	The substance which dissolves in a liquid to form a solution	The salt in seawater
Solution	The mixture formed when a solute is dissolved in a solvent	Seawater
Saturated solution	A solution with the maximum concentration of solute dissolved in the solvent	Seawater in the dead sea
Soluble	Describes a substance that will dissolve	Salt is soluble in water
Insoluble	Describes a substance that won't dissolve	Sand is insoluble in water
Filtrate	The liquid or solution that has passed through a filter	Fresh coffee in a cup
Residue	A substance that remains after evaporation, distillation, filtration or any other similar process	Coffee grounds in filter paper

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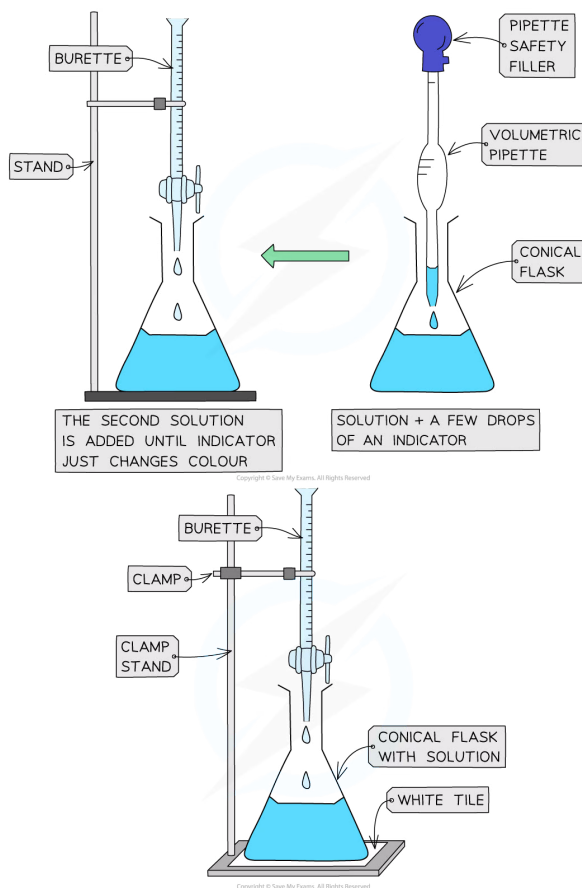
12.1.3 Acid-Base Titrations

Acid-Base Titrations

- Titrations are a method of analysing the **concentration** of solutions
- They can determine exactly how much alkali is needed to neutralise a quantity of acid – and vice versa
- You may be asked to calculate the **moles** present in a given amount, the **concentration** or **volume** required to **neutralise** an acid or a base
- Titrations can also be used to prepare **salts**

Describing how to carry out a titration

- **The typical materials needed are:**
 - 25 cm³ volumetric pipette
 - Pipette filler
 - 50 cm³ burette
 - 250 cm³ conical flask
 - Small funnel
 - 0.1 mol/dm³ sodium hydroxide solution
 - Sulfuric acid – concentration unknown
 - A suitable indicator
 - Clamp stand, clamp & white tile



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The steps in performing a titration

Describing the method:

1. Use the pipette and pipette filler and place exactly 25 cm^3 sodium hydroxide solution into the conical flask
2. Place the conical flask on a white tile so the tip of the burette is inside the flask
3. Add a few drops of a suitable indicator to the solution in the conical flask
4. Perform a rough titration by taking the burette reading and running in the solution in $1 - 3\text{ cm}^3$ portions, while swirling the flask vigorously
5. Quickly close the tap when the end-point is reached (sharp colour change) and record the volume, placing your eye level with the meniscus
6. Now repeat the titration with a fresh batch of sodium hydroxide
7. As the rough end-point volume is approached, add the solution from the burette one drop at a time until the indicator just changes colour
8. Record the volume to the nearest 0.05 cm^3
9. Repeat until you achieve two concordant results (two results that are within 0.1 cm^3 of each other) to increase accuracy

Indicators

- **Indicators** are used to show the endpoint in a titration
- Wide range indicators such as litmus are not suitable for titration as they do not give a sharp enough colour change at the endpoint
- However, methyl orange and phenolphthalein are very suitable
- Some of the most common indicators with their corresponding colours are shown below:

Common Acid-Base Indicators

Indicator	Colour in acid	Colour in alkaline	Colour in neutral
Litmus solution	Red	Blue	Purple
Litmus paper : red	Stays red	Turns blue	No change
Litmus paper : blue	Turns red	Stays blue	No change
Methyl orange	Red	Yellow	Orange
Phenolphthalein	Colourless	Pink	Colourless
Thymolphthalein	Colourless	Blue	Colourless

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12.2 Separation & Purification

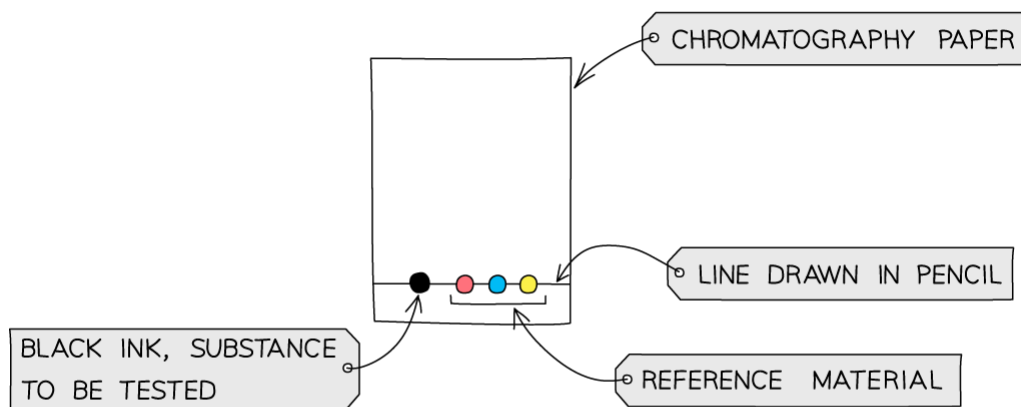
12.2.1 Paper Chromatography

Paper Chromatography

- This technique is used to separate substances that have **different solubilities** in a given solvent (e.g. different coloured inks that have been mixed to make black ink)
- A **pencil line** is drawn on chromatography paper and spots of the sample are placed on it. Pencil is used for this as ink would run into the chromatogram along with the samples
- The paper is then lowered into the solvent container, making sure that the pencil line sits **above** the level of the solvent so the samples don't wash into the solvent container
- The solvent travels up the paper by capillary action, taking some of the coloured substances with it
- Different substances have different solubilities so will travel at different rates, causing the substances to spread apart. Those substances with higher solubility will travel further than the others
- This will show the different components of the ink / dye

1

SET UP CHROMATOGRAPHY PAPER AS SHOWN



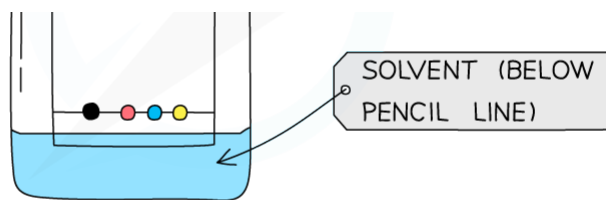
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LOWER PAPER INTO A BEAKER WITH APPROPRIATE SOLVENT. WAIT FOR SOLVENT TO TRAVEL UP THE PAPER.



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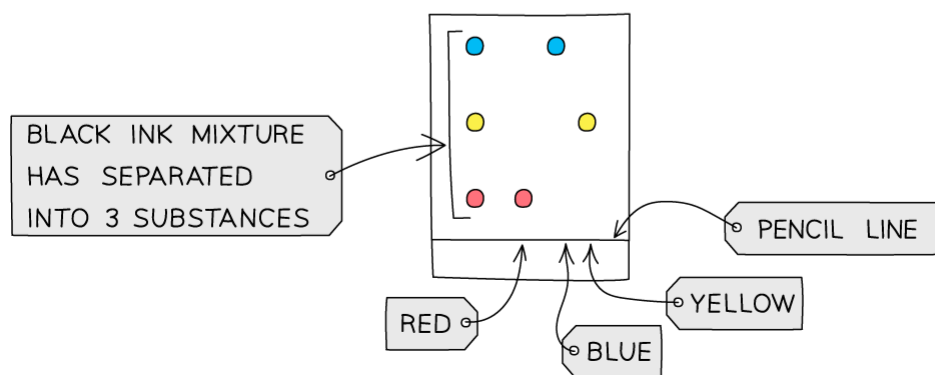


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ANALYSE CHROMATOGRAM



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Analysis of the composition of ink using paper chromatography

Interpret Simple Chromatograms

- If two or more substances are the same, they will produce identical chromatograms
- It is common practice to include a known compound as a reference spot
 - This can help match up to an unknown spot or set of spots in order to identify it
- If the substance is a mixture, it will separate on the paper to show all the different components as separate spots
- An impure substance will show up with more than one spot, a pure substance should only show up with one spot



Exam Tip

Paper chromatography is the name given to the **overall** separation technique while a **chromatogram** is the name given to the **visual output** of a chromatography run. This is the piece of chromatography paper with the visibly separated components after the run has finished.

12.2.2 Locating Agents & R_f Values**Locating Agents****EXTENDED**

- For chromatography to be useful the chemist needs to be able to see the components move up the paper, which is not the case for colourless substances such as amino acids or sugars
- Locating agents are substances which react with the sample and produce a coloured product which is then visible
- The chromatogram is treated with the agent **after** the chromatography run has been carried out, making the sample runs visible to the naked eye

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Retention Factor (R_f) Values

EXTENDED

- These values are used to identify the components of mixtures
- The R_f value of a particular compound is always the same
- Calculating the R_f value allows chemists to identify unknown substances because it can be compared with R_f values of known substances under the same conditions

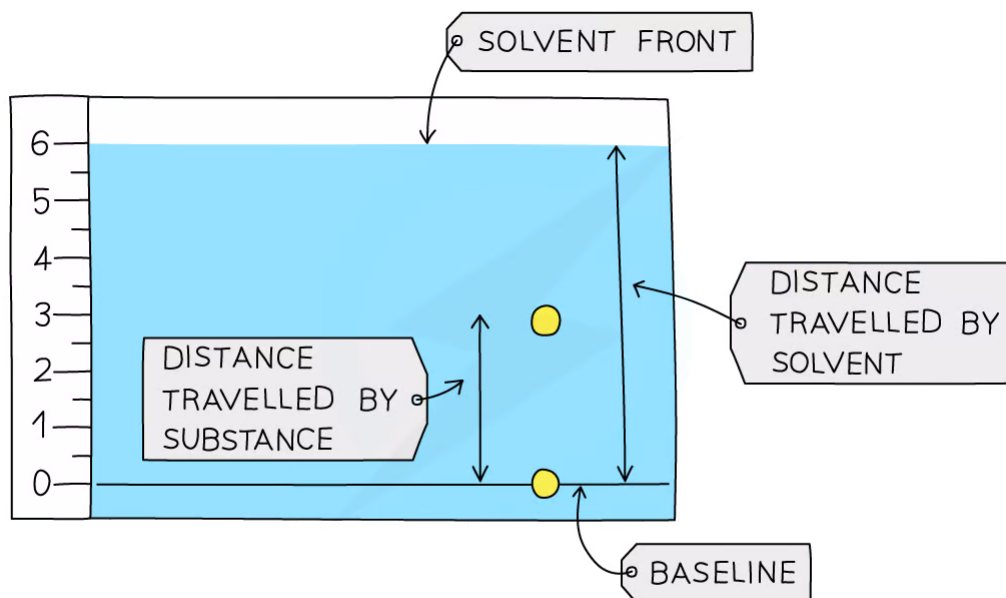
Calculation

- The formula to calculate the retention factor is:

$$\text{Retention factor} = \frac{\text{Distance travelled by substance}}{\text{Distance travelled by solvent}}$$

- The R_f value is a ratio and therefore has no units

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12.2.3 Separation & Purification Techniques

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Filtration & Crystallisation

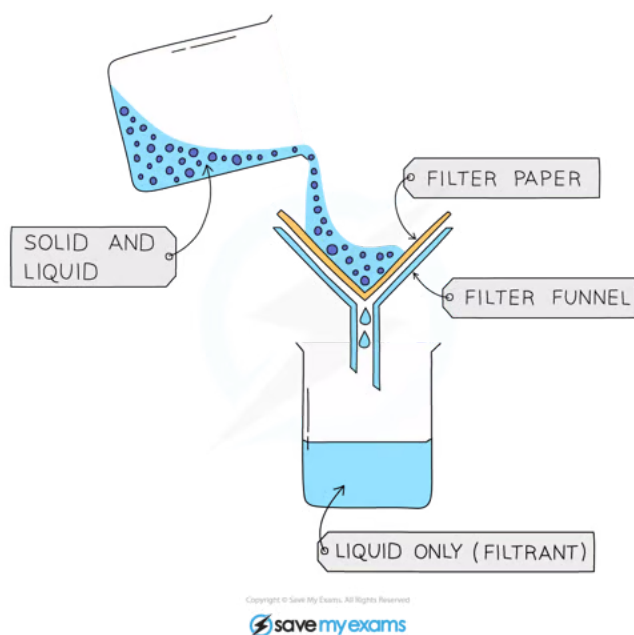
- The choice of the method of separation depends on the nature of the substances being separated
- All methods rely on there being a **difference** of some sort, usually in a physical property such as boiling point between the substances being separated

Separating a mixture of solids

- Differences in solubility can be used to separate solids
- For a difference in solubility, a suitable **solvent** must be chosen to ensure the desired substance only dissolves in it and not other substances or impurities, e.g. to separate a mixture of sand and salt, water is a suitable solvent to dissolve the salt, but not the sand

Filtration

- Used to separate an **undissolved solid** from a mixture of the solid and a liquid / solution (e.g. sand from a mixture of sand and water). Centrifugation can also be used for this mixture
- Filter paper is placed in a filter funnel above another beaker
- The mixture of insoluble solid and liquid is poured into the filter funnel
- Filter paper will only allow small liquid particles to pass through in the filtrate
- Solid particles are too large to pass through the filter paper so will stay behind as a residue



Filtration of a mixture of sand and water

Crystallisation

- Used to separate a **dissolved solid** from a solution, when the solid is more soluble in hot solvent than in cold (e.g. copper sulphate from a solution of copper (II) sulphate in water)
- The solution is heated, allowing the solvent to evaporate and leaving a saturated solution behind
- You can test if the solution is saturated by dipping a clean, dry, cold glass rod into the solution
 - If the solution is saturated, crystals will form on the glass rod when it is removed and allowed to cool
- The saturated solution is allowed to cool slowly and solids will come out of the solution as the solubility decreases, and crystals will grow
- Crystals are collected by filtering the solution
- They are then washed with distilled water to remove any impurities and then allowed to dry

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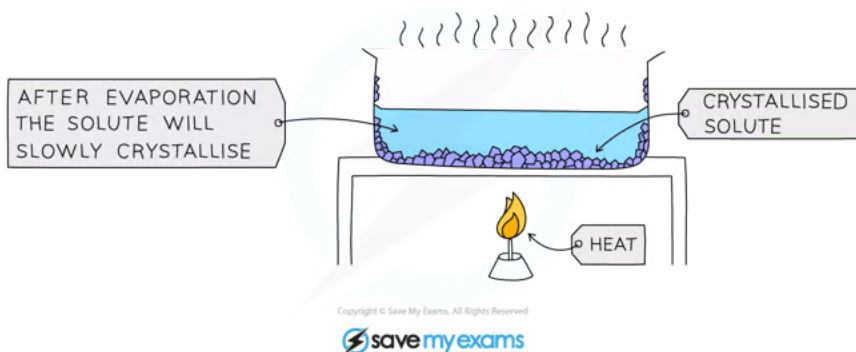


Diagram showing the process of crystallisation

Distillation: Simple & Fractional

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Simple Distillation

- Used to separate a liquid and **soluble solid** from a solution (e.g. water from a solution of saltwater) or a pure liquid from a mixture of liquids
- The solution is heated and pure water evaporates producing a vapour which rises through the neck of the round-bottomed flask
- The vapour passes through the condenser, where it cools and condenses, turning into pure water which is collected in a beaker
- After all the water is evaporated from the solution, only the solid solute will be left behind

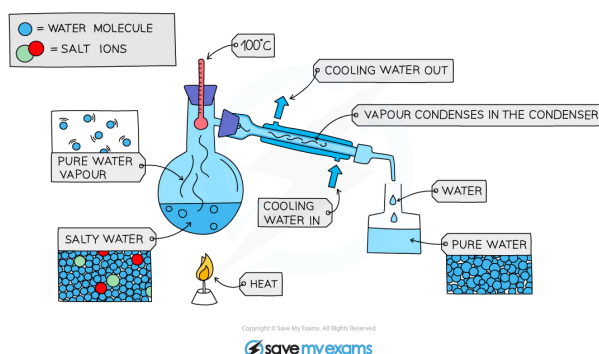
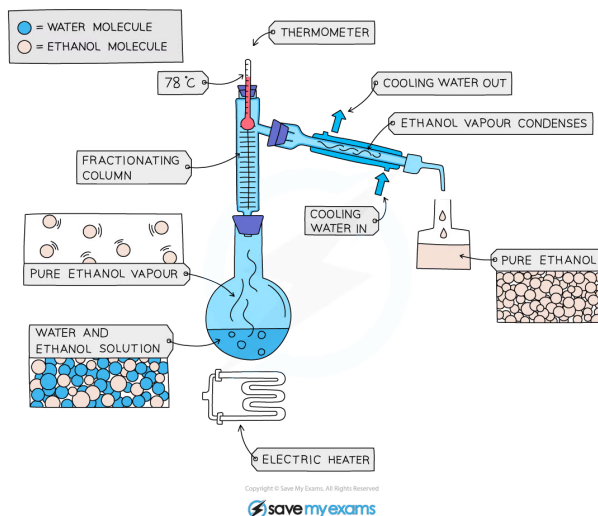


Diagram showing the distillation of a mixture of salt and water

- Simple distillation can be used to separate the products of fermentation, such as alcohol and water
- However, more effective separation is to use fractional distillation where the liquids are closer to boiling point and a higher degree of purity is required

Fractional distillation

- Used to separate two or more liquids that are miscible with one another (e.g. ethanol and water from a mixture of the two)
- The solution is heated to the temperature of the substance with the lowest boiling point
- This substance will rise and evaporate first, and vapours will pass through a condenser, where they cool and condense, turning into a liquid that will be collected in a beaker
- All of the substance is evaporated and collected, leaving behind the other component(s) of the mixture
- For water and ethanol: ethanol has a boiling point of 78 °C and water of 100 °C. The mixture is heated until it reaches 78 °C, at which point the ethanol boils and distills out of the mixture and condenses into the beaker
- When the temperature starts to increase to 100 °C heating should be stopped. Water and ethanol are now separated



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Fractional distillation of a mixture of ethanol and water

- An electric heater is safer to use when there are flammable liquids present
- The separation of the components in petroleum is achieved by fractional distillation on an industrial scale
- Fractional distillation of crude oil is not carried out in school laboratories due to the toxic nature of some of the components of the crude oil, but it can sometimes be simulated using a synthetic crude oil made specially for the demonstration



Worked Example

A student is given a mixture of calcium sulfate, magnesium chloride and water. The table below shows some information about calcium sulfate and magnesium chloride.

substance	solubility in water	state at room temperature
calcium sulfate	insoluble	solid
magnesium chloride	soluble	solid

How does the student obtain magnesium chloride crystals from the mixture?

- A crystallisation followed by distillation
- B crystallisation followed by filtration
- C distillation followed by crystallisation
- D filtration followed by crystallisation

Answer

The correct answer is **D** because:

- The difference in solubility in water means the first step is to make a solution

- The magnesium chloride will dissolve, but the solid calcium sulfate will be left behind
- The mixture is filtered to remove the calcium sulfate and then evaporated and crystallised to obtain magnesium chloride crystals



Exam Tip

You may be asked how to separate a mixture of gases. One method involves cooling the gaseous mixture sufficiently to liquefy all of the gases. The liquified mixture is then separated by fractional distillation. They can also be separated by diffusion, where the boiling points are very close or it is impractical or expensive to use fractional distillation.

Assessing Purity

- Pure substances melt and boil at specific and sharp temperatures. Eg water has a boiling point of 100°C and a melting point of 0°C
- Mixtures have a range of melting and boiling points as they consist of different substances that melt or boil at different temperatures
- Melting and boiling points data can therefore be used to distinguish pure substances from mixtures
- An unknown pure substance can be identified by experimentally determining its m.p and b.p and comparing to data tables
- Melting point analysis is routinely used to assess the purity of drugs for example
- This is done using a melting point apparatus which allows you to slowly heat up a small amount of the sample, making it easier to observe the exact melting point
- This is then compared to data tables
- The closer the measured value is to the actual melting or boiling point then the purer the sample is

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12.3 Identification of Ions & Gases

12.3.1 Identification of Anions

Identification of Anions

Anion	Test	Result
Carbonate (CO_3^{2-})	Add dilute acid and test the gas released	Effervescence, gas produced is CO_2 which turns limewater milky
Chloride (Cl^-)	Acidify with dilute nitric acid and add aqueous silver nitrate	White precipitate formed
Bromide (Br^-)	Acidify with dilute nitric acid and add aqueous silver nitrate	Cream precipitate formed
Iodide (I^-)	Acidify with dilute nitric acid and add aqueous silver nitrate	Yellow precipitate formed
Nitrate (NO_3)	Add aqueous NaOH and aluminium foil, warm gently and test the gas released	Gas given off is ammonia, has a pungent smell and turns moist red litmus paper blue
Sulfate (SO_4^{2-})	Acidify with dilute nitric acid and add aqueous barium nitrate	White precipitate formed
Sulfite (SO_3^{2-})	Add dilute acid, warm gently and test the gas released	Gas decolourises purple acidified aqueous potassium manganate(VII) solution

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Exam Tip

When it comes to qualitative inorganic analysis, always remember that there will be a test for the metal **cation** part of the molecule and another test for the **anion** part.

If you are an extended level student you may be asked to write balanced ionic equations for cation and anions tests, so make sure you know the formulae of all the ions and precipitates formed.

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12.3.2 Identification of Cations

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Identification of Cations

- Metal cations in aqueous solution can be identified by the **colour** of the precipitate they form on addition of sodium hydroxide and ammonia
- If only a small amount of NaOH is used then normally the resulting **metal hydroxide** precipitates out of solution
- In **excess** NaOH some of the precipitates may re-dissolve
- A few drops of NaOH is added at first and any colour changes or precipitates formed are noted
- Then the NaOH is added in excess and the reaction is observed again
- The steps are then repeated for the test using ammonia solution

Analysing results

- The table below contains the results for each of the cations included in the syllabus
- If a precipitate is formed from either NaOH or aqueous ammonia then it means that the hydroxide is insoluble in water
- Zinc chloride, for example, reacts as such:



- Ca^{2+} ions can be distinguished from Zn^{2+} and Al^{3+} as calcium hydroxide precipitate **does not dissolve** in excess NaOH but both zinc hydroxide and aluminium hydroxide do
- Zn^{2+} ions can be distinguished from Al^{3+} ions as $\text{Zn}(\text{OH})_2$ **dissolves** in excess aqueous ammonia but $\text{Al}(\text{OH})_3$ **does not**
- Most transition metals produce hydroxides with distinctive colours

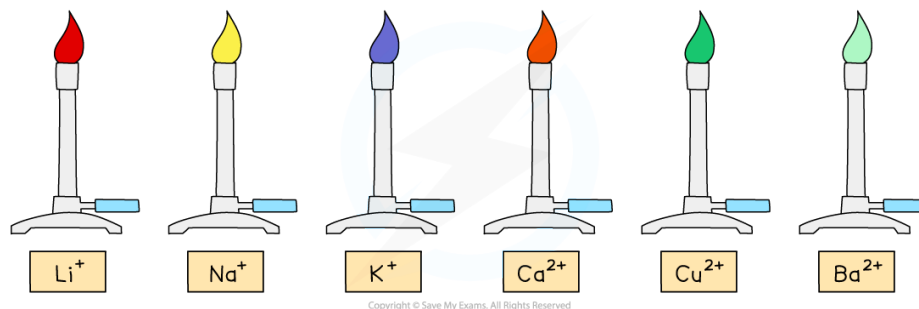
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METAL CATION	EFFECT OF ADDING NaOH	EFFECT OF ADDING AMMONIA SOLUTION
ALUMINIUM (Al^{3+})	WHITE PRECIPITATE, DISSOLVES IN EXCESS NaOH TO FORM A COLOURLESS SOLUTION	WHITE PRECIPITATE, INSOLUBLE IN EXCESS AMMONIA, WHITE PRECIPITATE REMAINS
AMMONIUM (NH_4^{+})	AMMONIA PRODUCED IF WARMED	—
CALCIUM (Ca^{2+})	WHITE PRECIPITATE, INSOLUBLE SO REMAINS IN EXCESS NaOH	VERY FAINTLY VISIBLE WHITE PRECIPITATE
CHROMIUM (III) (Cr^{3+})	GREEN PRECIPITATE WHICH FORMS A GREEN SOLUTION IN EXCESS	GREY-GREEN PRECIPITATE, INSOLUBLE IN EXCESS
COPPER (II) (Cu^{2+})	LIGHT BLUE PRECIPITATE, INSOLUBLE IN EXCESS	LIGHT BLUE PRECIPITATE, SOLUBLE IN EXCESS TO FORM DARK BLUE COLOUR
IRON (II) (Fe^{2+})	GREEN PRECIPITATE, INSOLUBLE IN EXCESS	GREEN PRECIPITATE, INSOLUBLE IN EXCESS
IRON (III) (Fe^{3+})	RED-BROWN PRECIPITATE, INSOLUBLE IN EXCESS	RED-BROWN PRECIPITATE, INSOLUBLE IN EXCESS
ZINC (Zn^{2+})	WHITE PRECIPITATE, DISSOLVES IN EXCESS TO FORM COLOURLESS SOLUTION	WHITE PRECIPITATE, DISSOLVES IN EXCESS TO FORM COLOURLESS SOLUTION

Tests for cations

- The flame test is used to identify the metal cations by the colour of the flame they produce
- A small sample of the compound is placed on an **unreactive** metal wire such as nichrome or platinum
- The colour of the flame is observed and used to identify the metal



The colours formed in the flame test for metal ions

Results

Cation	Colour of flame
Li^+	Red
Na^+	Yellow
K^+	Lilac
Ca^{2+}	Orange-red
Ba^{2+}	Apple-green
Cu^{2+}	Blue-green

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Exam Tip

The ammonia or sodium hydroxide solution must be added very slowly. If it is added too quickly and the precipitate is soluble in excess, then you run the risk of missing the formation of the initial precipitate, which dissolves as quickly as it forms if excess solution is added.

Be sure to distinguish between the term “**colourless**” and “**clear**”. A solution that loses its colour has become colourless. A clear solution is one that you can see through such as water. Solutions can be clear **and** have colour eg. dilute copper sulphate.

12.3.3 Identification of Gases

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Identification of Gases

- Several tests for anions and cations produce **gases** which then need to be tested
- The table below indicates the tests for the gases included in the syllabus

Gas	Appearance of gas	Test	Result
Ammonia (NH ₃)	Colourless, pungent smell	Damp red litmus paper	Turns blue
Carbon dioxide (CO ₂)	Colourless and odourless	Bubble through limewater	Limewater turns milky/cloudy
Chlorine (Cl ₂)	Pale green, choking smell	Damp blue litmus paper	Turns red and quickly is bleached
Hydrogen (H ₂)	Colourless and odourless	Hold a lighted splint in mouth of test tube	Burns with a 'squeaky pop' sound
Oxygen (O ₂)	Colourless and odourless	Hold a glowing splint	Splint relights
Sulfur dioxide (SO ₂)	Colourless, pungent choking smell	Add to acidified aqueous potassium manganate(VII)	Turns from purple to colourless

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Exam Tip

It is easy to confuse the tests for hydrogen and oxygen. Try to remember that a lig**H**ted splint has a **H** for **H**ydrogen, while a gl**O**wing splint has an **O** for **O**xygen.

